

CS 312: Operating System

Course Code	CS 312
Course Name	Operating System
L-T-P-C	3-0-0-3
Intended for	B.tech (CSE) 3rd Year, Electives for others
Approval	42nd Senate (Jan 2024), in the B.Tech CSE curriculum; 54th BoA (May 2024)

Course Contents

- **Introduction to Operating Systems (OS):** What OS do, OS Structure, OS Operations, OS Services, User and OS Interface, System Calls, Types of System Calls, System Programs, OS Design and Implementation, OS Debugging and OS Generations. (5 Hours)
- **Process Management:** Process Concept, Scheduling, Interprocess Communication, Multi-core and Multithreading Programming Models, Thread Libraries, Threading Issues, The Critical-Section Problem, Mutex Locks, Semaphores, CPU Scheduling, Scheduling Algorithms, Thread Scheduling, Deadlocks Characterization, Detection, Prevention and Avoidance. (9 Hours)
- **Memory Management:** Main Memory Background, Swapping, Contiguous Memory Allocation, Segmentation, Virtual Memory Concept, Demand Paging, Page Table Structure, Page Replacement, Copy-on-Write, Thrashing. (8 Hours)
- **Storage Management:** Storage Structure Overview, Disk Management and Scheduling, RAID Structure, File System Interface, Directory and Disk Structure, File Sharing, File-System Structure and Allocation Methods, Directory Implementation, I/O Hardware, I/O Interface, Kernel I/O Subsystem. (8 Hours)
- **Protection and Security:** Goals, Principle and Domain of Protection, Access Matrix. Access Control, Access Rights Revocation, Security Problem, Program Threats, User Authentication, Implementing Security Defenses, Firewalls. (7 Hours)
- **Advance Topics:** Virtual Machines, Virtualization and OS Component, Distributed System, Network-based OS, Communication Protocols, Case Studies related to different OS like XV6, Linux, Windows, etc. (5 Hours)

Text Books

1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, **Operating System Concepts**, 8th Edition, John Wiley, 2008.
2. AS Tanenbaum, **Modern Operating Systems**, 3rd Edition, Pearson, 2009

References

1. William Stallings, **Operating Systems: Internals and Design Principles**, Prentice-Hall, 6th Edition, 2008.
2. AS Tanenbaum, AS Woodhull, **Operating Systems Design and Implementation**, 3rd Edition, Prentice Hall, 2006.
3. M. J. Bach, **Design of the Unix Operating System**, Prentice Hall of India, 1986.
4. Remzi Arpaci-Dusseau, Andrea Arpaci-Dusseau, **Operating Systems: Three Easy Pieces**, Version 1.1.0.